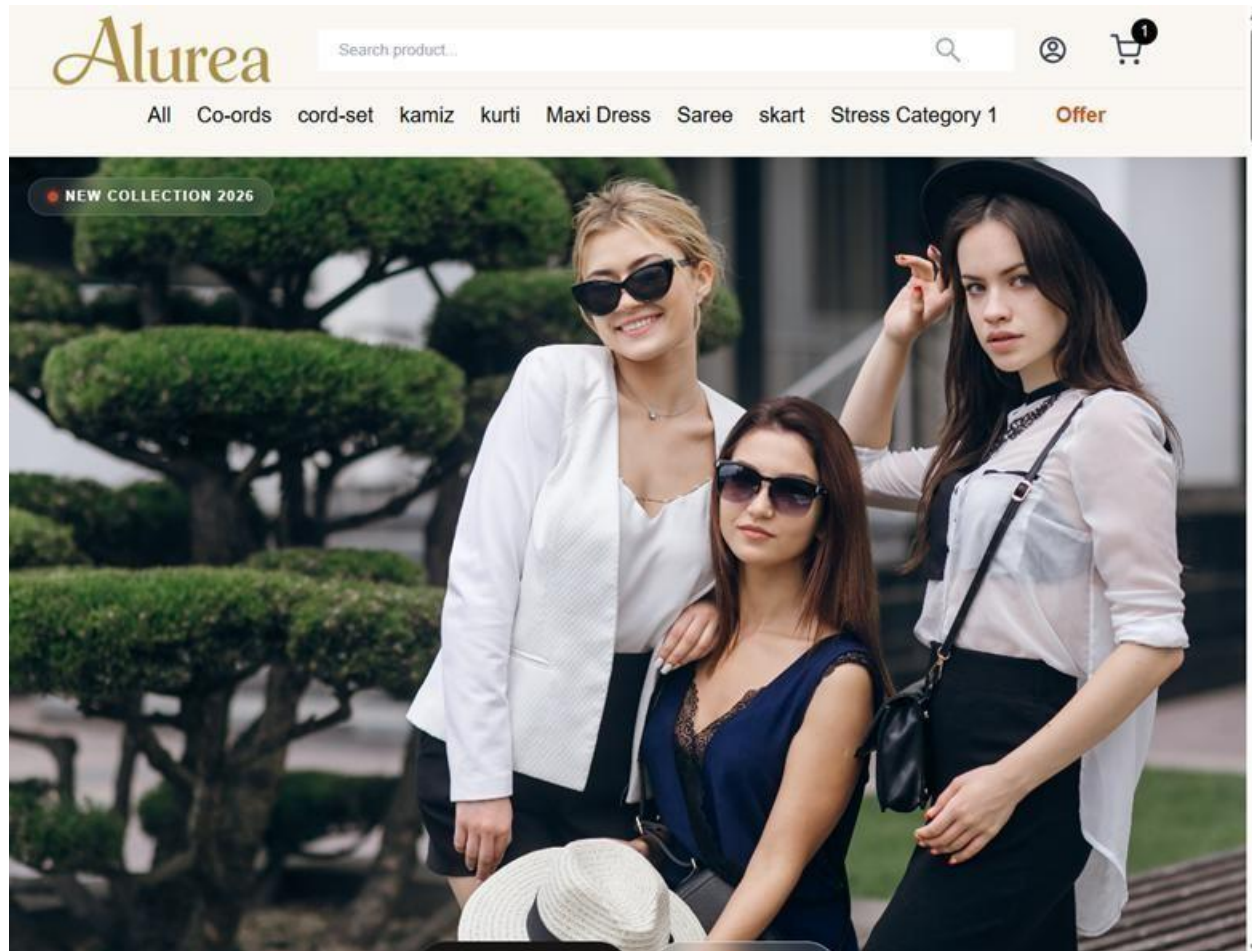


# Complete eCommerce Business Management System



## Overview

A scalable full-stack eCommerce Business Management System built to help businesses manage products, inventory, customer orders, payments, courier operations, and business analytics from a centralized admin dashboard.

The platform provides customers with a seamless shopping experience while giving administrators complete control over daily business operations through a secure and responsive management system.

## Tech Stack

### Frontend

- Next.js 15
- React

- TypeScript
- Tailwind CSS

## Backend

- NestJS
- PostgreSQL
- JWT Authentication
- Cloudinary

## Project Links

**Live Demo:** <https://www.sumashop.xyz/>

## Key Features

### Customer Website

- Responsive Shopping Experience
- Product Search
- Product Variants
- Shopping Cart
- Secure Checkout
- Fetches real-time shipping rates via Shippo API
- Customers can compare available shipping options
- Customers can select their preferred shipping rate/service during checkout.
- Cash on Delivery
- Stripe Mock Payment
- Order Tracking
- Dark / Light Mode

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### Admin Panel

- Product Management
- Category Management
- Variant Management
- Inventory Management
- Order Management

- Customer Management
- Payment Verification
- Courier Dispatch
- Sales Dashboard

## **Advanced Features**

### **Fraud Detection System**

- Risk Score Calculation
- Fake COD Detection
- Suspicious Address Detection
- Repeated Phone Detection
- Same IP Detection
- High Value COD Detection

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### **Smart Inventory**

- Automatic Stock Reservation
- Automatic Stock Restore
- Variant Stock Management

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### **Courier Integration**

- Shippo API
- Shipment Tracking
- Tracking Link

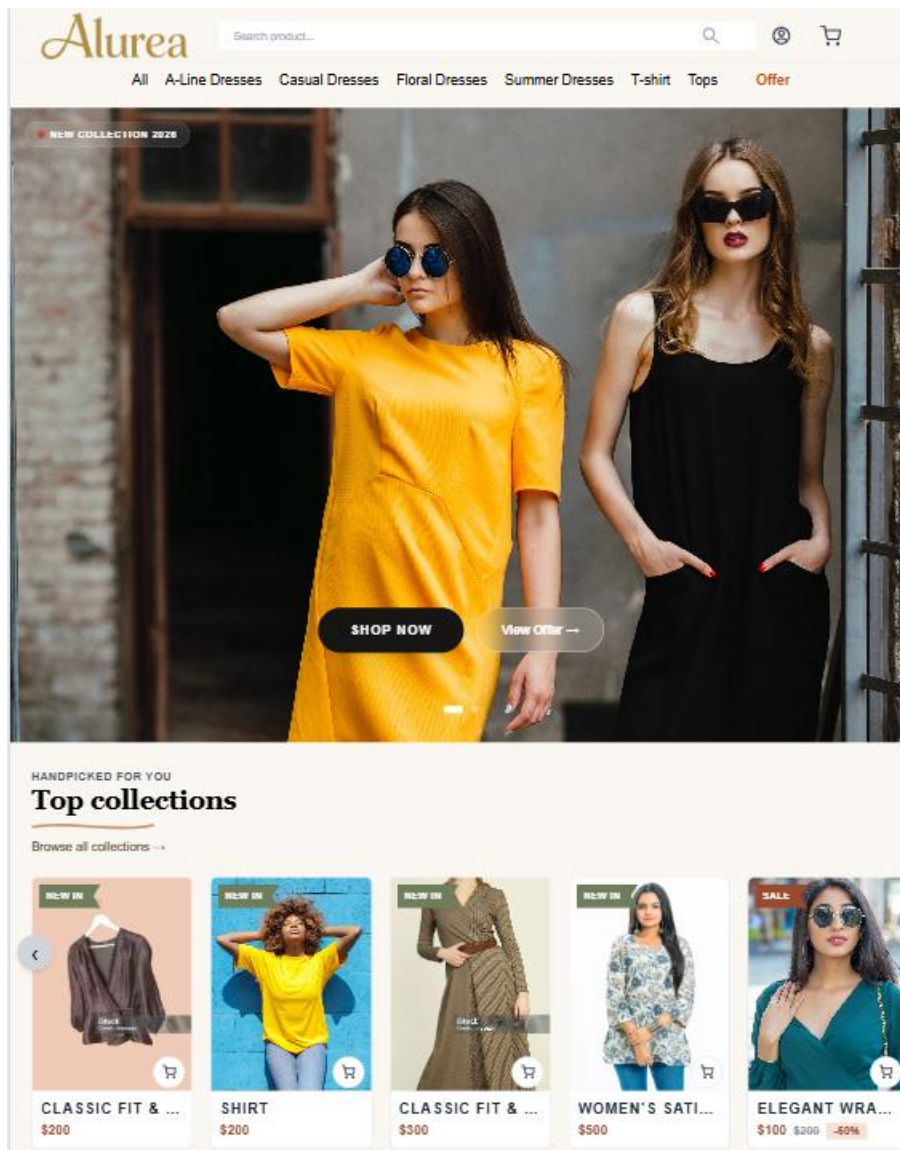


Figure 1: Alurea — Full-Stack E-commerce Storefront Homepage



## Rosewood Floral Tiered Dress

**\$400** ~~\$500~~

select color



select size

36 38 40

- 1 +

Add to cart

Buy Now

- ✓ 100% Original Product
- 🚚 Express Shipping
- 💳 Cash on Delivery Available
- 🔄 Easy return and exchange within 3 days

### Product Description

Step into effortless elegance with the Ruby Bloom Floral Maxi Dress, designed with a beautiful floral print and a graceful, flowing silhouette. The flattering cinched waist, long puff sleeves, and tiered hem

Figure 2: Alurea Product Detail Interface Featuring Product Gallery, Variants, and Add-to-Cart Workflow

Alurea

Search product...

Checkout

Full Name

Atik

Email

sumafatema10@gmail.com

Phone

+1xxxxxxxxxx

City

San Francisco

State

CA

Postal Code

94103

Country

United States

Address

Enter your full address

Shipping Method

Get Shipping Rates

Payment Method

☒ Cash on Delivery

☐ Card Payment — Stripe Unavailable

Order Summary

 Women's Satin Evening...  
Qty: 1

\$500

Subtotal

\$500

Delivery

\$0

Total

\$500

Place Order

Figure 3:Alurea Checkout Page with Shipping, Payment, and Order Summary Workflow

# All Orders

Check all orders at single place.

All Order (7) Pending (0) Processing (1) Shipped (1) Completed (3) Cancelled (0)

|                                                                                                                                       |                                                                             |           |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------|
|                                                      | <div>Classic Fit &amp; Flare Dress</div> <div>\$200</div> <div>Qty: 1</div> | completed |
| <div>Courier: Shippo</div> <div>Status: delivered</div> <div>Tracking: 9300120845500000537341</div> <div>Track Parcel</div>           |                                                                             |           |
|                                                      | <div>shirt</div> <div>\$200</div> <div>Qty: 1</div>                         | completed |
| <div>Courier: Shippo</div> <div>Status: assigned to courier</div> <div>Tracking: 9300120845500000537358</div> <div>Track Parcel</div> |                                                                             |           |

Figure 4: Alurea Order Dashboard Featuring Order Status, Courier Tracking, and Delivery Updates

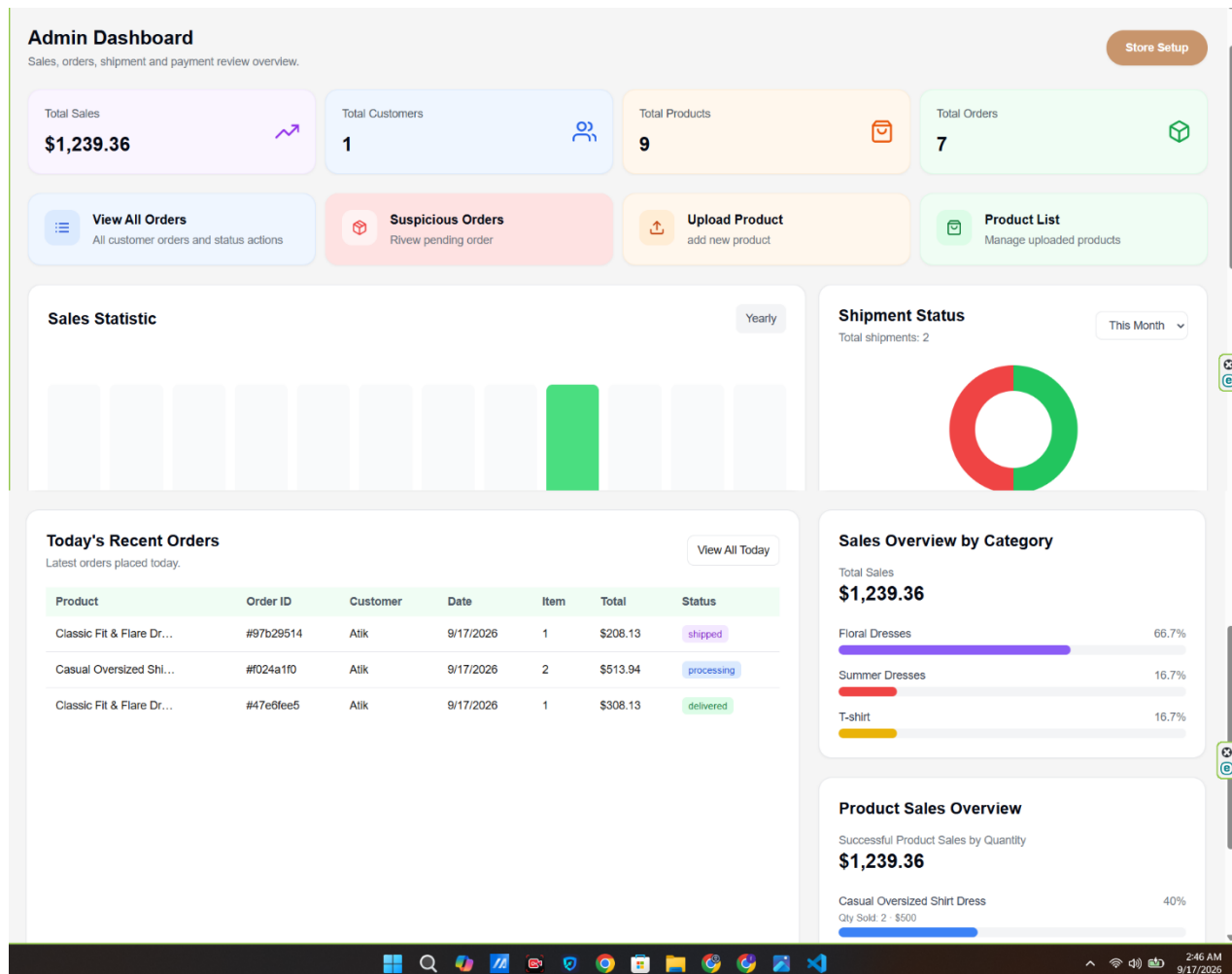


Figure 5: Alurea Admin Management Dashboard Featuring Sales Analytics, Order Monitoring, Product Management, and Shipment Overview

A

Admin Panel

Management Dashboard

Hello, user1 🧑

U

user1

Administrator

Products

Active products: 1065

+ Upload Product

Search by product name or slug

All Products

Active Products

All Categories

Clear Filter

| Product                                                                                                                                                                                                           | Category           | SKU                      | Price                                   | Stock | Status                                     | Actions                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------------------------------|-------|--------------------------------------------|----------------------------------------------------------------------|
| <div></div> <div><div>Rosewood Floral Tiered Dress</div><div>/rosewood-floral-tiered-dress</div><div>Variant Product</div></div> | Saree              | PRD-1787413376679-434151 | <div>\$400.00</div> <div>\$500.00</div> | 36    | <div>Published</div> <div>Flash Deal</div> | <div>Edit</div> <div>Archive</div> <div>Delete</div> <div>View</div> |
| <div></div> <div><div>Lightweight, flowy silhouette</div><div>/saree-9</div></div>                                               | Stress Category 18 | PRD-1785693078521-6B610C | <div>\$788.00</div>                     | 5     | <div>Published</div>                       | <div>Edit</div> <div>Archive</div> <div>Delete</div> <div>View</div> |
| <div></div> <div><div>Crimson Bloom Maxi Dress</div><div>/saree-8</div><div>Variant Product</div></div>                          | Stress Category 17 | PRD-1785692385298-B98A78 | <div>\$300.00</div> <div>\$500.00</div> | 8     | <div>Published</div> <div>Flash Deal</div> | <div>Edit</div> <div>Archive</div> <div>Delete</div> <div>View</div> |
| <div></div> <div><div>Rosewood Floral Tiered Dress</div><div>/top1-2</div></div>                                                | Stress Category 16 | PRD-1785468945371-0A84A8 | <div>\$700.00</div>                     | 4     | <div>Published</div>                       | <div>Edit</div> <div>Archive</div> <div>Delete</div> <div>View</div> |

Figure 6: Alurea Admin Product Management Dashboard

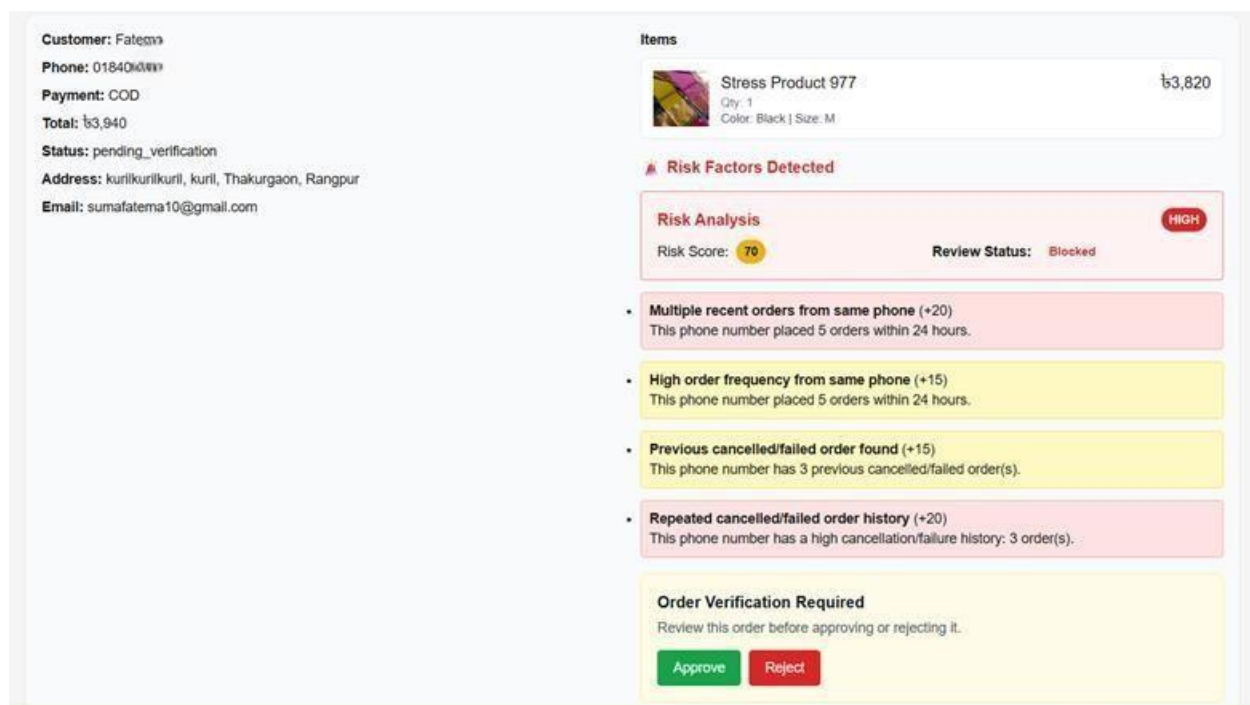


Figure 7: Automated order risk analysis dashboard identifying suspicious orders through customer behaviour patterns and verification checks.

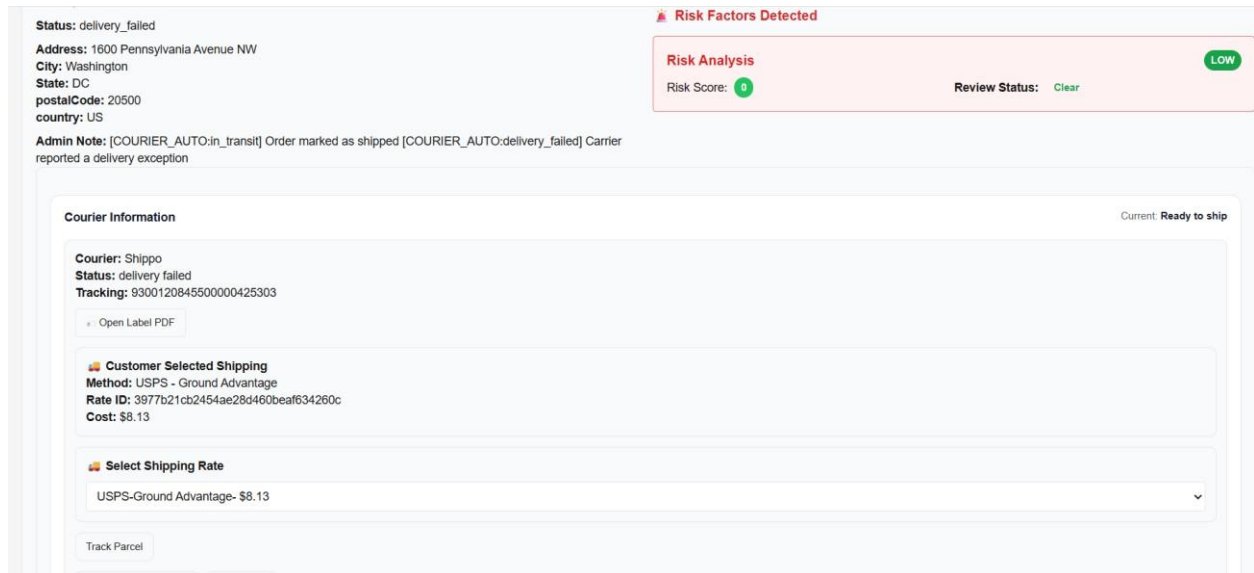


Figure 8: Shipment management interface displaying courier details, tracking information, shipping rates, and delivery status updates.

## Technical Architecture

### Frontend

- Next.js 15
- TypeScript
- Tailwind CSS

### Backend

- NestJS
- PostgreSQL

### Authentication

- JWT
- RBAC

### Storage

- Cloudinary

## Deployment

- Vercel
- Render

# Business Value

Why this project?

- Complete business management
- Reduce fake orders
- Save courier costs
- Improve inventory accuracy
- Increase operational efficiency
- Better customer experience

## Project Statistics

- ✓ 50+ Business Features
- ✓ 27+ Admin Features
- ✓ 21+ Customer Features
- ✓ 11 Order Statuses
- ✓ 7 Fraud Detection Rules
- ✓ 2 Payment Methods
- ✓ Shippo API Integration
- ✓ JWT Authentication & RBAC
- ✓ Responsive Design
- ✓ Light & Dark Mode

## Contact



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Live Demo <https://sumashop.xyz/>

Github:

<https://github.com/Fatema-Akther/ecommerce-platform-backend>

<https://github.com/Fatema-Akther/ecommerce-platform-Frontend>

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